$$\langle \sigma_A \sigma_B \rangle = -1.$$

Notice that the average of the products (of σ_A and σ_B)is not equal to the product of the averages—Eqs. 6.1 tell us that $<\sigma_A><\sigma_B>$ is zero. In symbols,

$$\langle \sigma_A \rangle \langle \sigma_B \rangle \neq \langle \sigma_A \sigma_B \rangle,$$

or

$$\langle \sigma_A \sigma_B \rangle - \langle \sigma_A \rangle \langle \sigma_B \rangle \neq 0. \qquad (6.2)$$

This indicates that Alice's and Bob's observations are *correlated*. In fact, the quantity

$$\langle \sigma_A \sigma_B \rangle - \langle \sigma_A \rangle \langle \sigma_B \rangle$$

is called the statistical correlation between Bob's and Alice's observations. It's called the statistical correlation even if it is zero. When the statistical correlation is nonzero, we say the observations are correlated. The source of this correlation is the fact that originally Alice and Bob were in the same location and Charlie had one of each type of coin. The correlation remained when Alice went to Alpha Centauri simply because the coins didn't change during the trip. There is absolutely nothing strange about this or about Inequality 6.2. It is a very common property of statistical distributions.

Suppose you have a probability distribution $P(a, b)$ for two variables a and b. If the variables are completely uncorrelated, then the probability will factorize:

$$P(a, b) = P_A(a) P_B(b), \qquad (6.3)$$